

Cochlear Implant

ACG: A-0177 (AC)

[Link to Codes](#)

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Clinical Indications

- Cochlear implant may be indicated for **1 or more** of the following(1)(2)(3)(4)(5):
 - ☐ Adult[A] with **ALL** of the following(14)(15)(16):[N](#)
 - Intact cochlear nerves confirmed by CT or MRI, or acoustic neuroma (vestibular schwannoma) excision planned and cochlear nerve preservation thought possible(33)(34)
 - Need for implant, as indicated by **1 or more** of the following:
 - Asymmetric sensorineural hearing loss and **ALL** of the following(26)(27)(28)(29)(35):
 - Hearing loss, unaided, of 70 dB or greater (ie, severe to profound hearing loss) in one ear
 - Contralateral ear with 31 to 55 dB hearing loss (ie, mild to moderately severe hearing loss), with difference of at least 15 dB in pure-tone averages between ears
 - Limited benefit from appropriately fitted hearing aid in ear to be implanted
 - Bilateral sensorineural hearing loss of greater than 40 dB[B] with limited (eg, best aided word or open-set sentence cognition test at 60% or less, presented at 60 dB) speech perception benefit from hearing aids(20)(36)(37)
 - Unilateral sensorineural hearing loss and **ALL** of the following(26)(29)(35):
 - Hearing loss, unaided, of 70 dB or greater (ie, severe to profound hearing loss) in one ear
 - Contralateral ear hearing loss less than 30 dB (ie, normal or mild hearing loss)
 - Limited benefit from appropriately fitted hearing aid in ear to be implanted
 - Unilateral sensorineural hearing loss and risk for progression (eg, due to bacterial meningitis)(30)
 - No lesions of acoustic nerve or central auditory pathway causing deafness
 - ☐ Child[A] with **ALL** of the following(38)(39)(40)(41)(42):[N](#)
 - Cochleovestibular anatomy compatible with cochlear implant placement (ie, no evidence of anomaly that would preclude implant, such as cochlear aplasia, complete labyrinthine aplasia, lack of cochlear nerve), confirmed by CT or MRI, or acoustic neuroma (vestibular schwannoma) excision planned and cochlear nerve preservation thought possible(31)(54)(57)(58)(59)
 - Need for implant, as indicated by **1 or more** of the following:
 - Asymmetric sensorineural hearing loss and **ALL** of the following(35):
 - Hearing loss, unaided, of 70 dB or greater (ie, severe to profound hearing loss) in one ear
 - Contralateral ear with 31 to 55 dB hearing loss (ie, mild to moderately severe hearing loss), with difference of at least 15 dB in pure-tone averages between ears
 - Limited benefit from appropriately fitted hearing aid in ear to be implanted

- Bilateral sensorineural hearing loss, as indicated by **ALL** of the following:
 - Age 9 months or older^(C)(45)(49)(50)(62)(63)
 - Hearing loss with unaided pure-tone average thresholds of 70 dB or greater (ie, severe hearing loss)^(B)
 - Three-month to six-month trial of binaural hearing aids documents lack of or minimal improvement (ie, less than appropriate based on age, developmental stage, or cognitive ability) in auditory development.^(D)
- Unilateral sensorineural hearing loss and **ALL** of the following⁽³¹⁾⁽³⁵⁾⁽⁵²⁾⁽⁵⁵⁾:
 - Hearing loss, unaided, of 70 dB or greater (ie, severe to profound hearing loss) in one ear
 - Contralateral ear with less than 30 dB hearing loss (ie, normal to mild hearing loss), with difference of at least 15 dB in pure-tone averages between ears
 - Limited benefit from appropriately fitted hearing aid in ear to be implanted
- Unilateral sensorineural hearing loss and risk for progression (eg, due to bacterial meningitis)⁽³¹⁾
 - Family support and motivation to participate in postimplant rehabilitation⁽⁵⁰⁾⁽⁵⁷⁾
 - Minimal speech perception 30% or less or lack of developmentally appropriate auditory milestones measured using parent report scales
 - No evidence of central auditory dysfunction (eg, cortical deafness)
- Replacement of existing cochlear implant or component (eg, headset/headpiece, microphone, transmitting coil or cable, external speech processor or controller, battery) needed⁽¹⁰⁾^(N)

Alternatives

- Alternatives include⁽³⁹⁾⁽⁴¹⁾:
 - Auditory brainstem implant, when cochlear nerve not intact. See Auditory Brainstem Implants ^(N) AC for further information.
 - Bone anchored hearing aid. See Hearing Aids, Bone Anchored and Bone Conduction ^(N) AC for further information.
 - Hearing aid⁽¹⁶⁾

Evidence Summary

Background

A cochlear implant transforms sound into electrical energy that stimulates surviving auditory nerve fibers in the inner ear. All cochlear implant devices consist of internal and external hardware. External components are worn either behind the ear or on the body. They include a microphone, a sound processor, batteries, and a transmitter that send coded electrical information and power to the internal parts. The internal components are placed surgically underneath the skin and include a receiver and an electrode array within the cochlea.⁽¹⁾⁽⁶⁾⁽⁷⁾⁽⁸⁾⁽⁹⁾ **(EG 2)**

Criteria

The evidence for the clinical indications found in this guideline includes 57 published peer reviewed articles, 2 specialty society or other evidence-based guidelines, and 3 book sections.

For adults with hearing loss, evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** A systematic review and meta-analysis of 14 studies (679 adult patients) evaluating quality-of-life improvement after cochlear implantation found that cochlear implantation was associated with significant improvement in quality of life as measured by hearing-specific or cochlear implant-specific quality-of-life patient-reported outcomes.⁽¹⁷⁾ **(EG 1)** A systematic review of 3 randomized controlled trials and 7 observational studies including 308 adult patients with severe to profound sensorineural hearing loss found that, compared with unilateral cochlear implantation, bilateral cochlear implantation was associated with improved speech perception in noise, sound localization, and subjective improvements in speech and spatial hearing.⁽²⁾ **(EG 1)** A systematic review of unilateral vs bilateral cochlear implantation in adults concluded that unilateral implantation with or without the use of hearing aids was effective for improving speech perception in adults with severe to profound sensorineural hearing loss; both simultaneous and sequential bilateral cochlear implantation provided additional improvement in speech perception.⁽¹⁸⁾ **(EG 1)** An industry-sponsored randomized controlled trial of 38 adults with postlingual, severe to profound hearing loss compared simultaneous and sequential (2 years between procedures) bilateral cochlear implants and found, 1 year after both implants were in place, comparable results between the groups in terms of speech intelligibility in noise from straight ahead, from spatially separated sources, and in silence. The authors concluded that patients who receive sequential implants derive the same benefit as those who receive them simultaneously.⁽¹⁹⁾ **(EG 1)** A national guideline recommends simultaneous bilateral cochlear implantation only for adults with severe to profound deafness who are blind or who have other disabilities that increase their reliance on auditory stimuli as a primary sensory mechanism of spatial awareness.⁽³⁾ **(EG 2)** A cross-sectional study of 323 adult patients with bilateral sensorineural hearing loss (470 cochlear implants) evaluated the effectiveness of treatment with cochlear implantation as measured by preoperative and postoperative word recognition, sentence recognition in quiet, and sentence recognition in noise scores (as measured by consonant-nucleus-consonant word test, AzBio sentences in quiet, and AzBio sentences in +10-dB signal to noise ratio, respectively) at 60 dB and found, at 6-month to 12-month follow-up, that cochlear implantation was associated with improved postoperative word, sentence in quiet, and sentence in noise recognition in 85%, 88%, and 79% of patients, respectively. The findings were limited by study methodology; further well-designed studies were recommended.⁽²⁰⁾

(**EG 2**) A prospective study of 94 postlingual deaf patients (65 to 85 years of age) who were treated with cochlear implants for sensorineural hearing loss found a mean improvement in speech perception scores of 52% at 6 months, with continued improvement at 12 months; there was also significant improvement in quality of life. Patients with depression, as assessed by the Geriatric Depression Scale-4 (GDS-4), decreased from 41% to 24% at 12 months after implantation.(21) (**EG 2**) A systematic review and meta-analysis of 29 studies (1868 patients) comparing the magnitude of improvement in word and sentence recognition scores between younger (younger than 60 years) and older (older than 60 years) adults after cochlear implantation found that there was no difference in the preoperative and postoperative effect sizes in word and speech recognition scores between age groups.(22) (**EG 1**) A systematic review and meta-analysis of 36 studies (796 patients) evaluating cochlear implantation in adult patients with single-sided deafness (defined as pure-tone average of 70 dB or greater in the worse-hearing ear, with pure-tone average less than 35 dB in the contralateral ear) found that cochlear implantation was associated with improved speech recognition in quiet and noise, sound localization, and patient-reported tinnitus severity and quality of life.(23) (**EG 1**) A case series of 12 patients and a systematic review of 20 studies including 97 patients (10 studies with 66 patients included in meta-analysis) evaluating the efficacy of cochlear implants in patients with radiation therapy-induced sensorineural hearing loss found that, while implantation is feasible and results show mean improvement in speech discrimination scores of 56.2%, abnormal intraoperative findings due to underlying pathology and/or cancer treatment and postoperative complications are common in this population.(24) (**EG 1**) A systematic review and meta-analysis of 52 studies (201 postlingually deafened adult patients) evaluating the effect of cochlear implantation on cognitive measures found a positive association of cochlear implantation in 50.8% of outcomes measured overall, with the greatest percentage in memory and learning (60.5%) and lowest in construct verbal fluency (17.6%). The authors noted that differences in baseline cognitive function of patients, variations among cognitive tests used, and heterogeneity in patient demographics limited the results.(25) (**EG 1**) A prospective repeated-measures study of 47 adults with postlingual asymmetric hearing loss (severe to profound hearing impairment in the poorer ear and normal to moderately severe hearing impairment in the better ear) evaluated treatment of the poorer ear with cochlear implant and found, at 6-month follow-up, improved audibility and speech recognition tests (eg, consonant-vowel nucleus-consonant and Bamford-Kowal-Bench sentence in noise). The findings were limited by small study numbers and short follow-up; further well-designed trials with larger study populations and longer follow-up were recommended.(26) (**EG 2**) A prospective repeated-measures study of 44 adult patients with asymmetric hearing loss (defined as audiometric hearing loss between 31 dB and 60 dB sound pressure level at up to 4000 Hz and speech discrimination score greater than 50% on word testing measured at 65 dB sound pressure level in the better ear) found, at 6-month and 12-month follow-up, that cochlear implantation was associated with improved speech discrimination scores (as measured by Freiburg Monosyllabic Word Test and Oldenburg Sentence Test at 65 dB, subjective hearing testing in quiet, hearing in noise, and sound localization), health-related quality of life, and tinnitus distress. There was no difference in stress level, depressiveness, or general anxiety, but coping with stress improved (as measured by Perceived Stress Questionnaire, General Depression Scale, Generalized Anxiety Disorder-7, and COPE Inventory, respectively).(27) (**EG 2**) A randomized controlled trial of 51 adult patients with single-sided deafness or asymmetric hearing loss, who failed a successive 3-week trial each of contralateral routing of signal (CROS) hearing aid system and bone conduction headband devices from a previous observational cohort study, compared subsequent cochlear implantation vs observation and found, at 6-month follow-up, that cochlear implantation was associated with improved tinnitus severity and quality of auditory perception (as measured by visual analog scale and Nijmegen Cochlear Implant Questionnaire, respectively); there was no difference in quality of life (as measured by EuroQol EQ-5D) between groups. The authors noted that the findings were limited by small sample size and short-term follow-up; further well-designed studies with larger study populations and longer follow-up were recommended.(28)(29) (**EG 1**) A technology assessment found moderate-quality evidence that cochlear implants improve sound localization, speech perception in noise, tinnitus symptoms, and quality of life in adults and children with single-sided deafness or asymmetric hearing loss.(5) (**EG 2**) Consensus guidelines recommend consideration of early cochlear implantation for adults and children with single-sided hearing loss who are at risk for further rapid progressive hearing loss due to ossification (eg, after bacterial meningitis, acoustic neuroma (vestibular schwannoma) resection, otic capsule fracture), given the risk to future successful cochlear implantation.(30)(31) (**EG 2**) An industry-affiliated task force recommends evaluation for cochlear implantation in patients with pure-tone average greater than or equal to 60 dB and word recognition scores less than or equal to 60% in at least one ear.(32) (**EG 2**)

For infants or children with hearing loss, evidence demonstrates at least moderate certainty of at least moderate net benefit. (**RG A1**) A systematic review of prognostic factors for cochlear implant in children found that improved outcomes were associated with early implant, congenital deafness due to GJB-2 gene mutation, less severe inner ear malformations, and early implant of postmeningitic or congenitally deaf children.(43) (**EG 1**) Multiple studies of unilateral cochlear implant in children demonstrate that functional outcomes are improved when the surgery is performed at a younger age. Eligible children should receive a cochlear implant as soon as bilateral profound hearing impairment is diagnosed to maximize speech and language achievement and integration into an oral communication environment. Children who are implanted when younger than 2 years can experience normal or near-normal rates of auditory skill and oral language development. However, even in older children, the oral language and speech benefits of implant are substantial for those who have some residual hearing because they are able to hear more speech and sound information with the cochlear implant than with a hearing aid.(44)(45)(46)(47)(48) (**EG 2**) A systematic review of 14 studies evaluating the effect of early (before age 12 months) cochlear implantation found better scores on speech production, auditory performance, and some receptive language tests in children implanted before age 12 months compared with those implanted later. However, the authors noted that the available evidence consisted of cohort studies with moderate to high risk of bias and recommended long-term follow-up studies.(49) (**EG 1**) A systematic review of 4 studies with a total of 103 pediatric patients found that simultaneous bilateral implantation, as compared with sequential bilateral implantation, resulted in statistically significant higher speech and language development scores 3 years after the first cochlear implantation.(12) (**EG 1**) Children with bilateral cochlear implants that are activated at earlier ages and with shorter gaps between surgeries appear to receive greater benefit than those implanted later and with longer gaps between surgeries.(50) (**EG 2**) Other systematic reviews that compared bilateral cochlear implant with unilateral implant in children found that, although the data are limited, bilateral cochlear implant appeared to be more effective in terms of sound localization and improved speech perception in quiet and

noise.(13)(51) **(EG 1)** A systematic review and meta-analysis of 12 observational studies with 119 pediatric patients (mean age 6.6 years) with single-sided deafness (unaided pure-tone average of 90 dB or greater in one ear) found that cochlear implants improved speech perception in noise in 79.6% of patients and speech perception in quiet in 81% of patients; cochlear implants were also associated with improved sound localization.(52) **(EG 1)** A systematic review of 8 studies (390 children with residual hearing) comparing cochlear implantation with conventional hearing aids found that cochlear implantation was associated with improved speech perception scores compared with conventional hearing aids; partial or total loss of residual hearing occurred in 37.2% of patients after cochlear implantation. The authors noted that variability in participant characteristics and candidacy criteria for implantation, duration of device use, and measures used in the studies limited the results.(53) **(EG 1)** A technology assessment found moderate-quality evidence that cochlear implants improve sound localization, speech perception in noise, tinnitus symptoms, speech and language development, and quality of life in children with single-sided deafness or asymmetric hearing loss.(5) **(EG 2)** Review articles support cochlear implantation as early as possible (ie, within the first 5 years of life) in pediatric patients with single-sided deafness to help prevent reorganization of central auditory pathways and improve binaural hearing outcomes.(54)(55)(56) **(EG 2)** Consensus guidelines recommend consideration of early cochlear implantation for adults and children with single-sided hearing loss who are at risk for further rapid progressive hearing loss due to ossification (eg, after bacterial meningitis, acoustic neuroma (vestibular schwannoma) resection, otic capsule fracture), given the risk to future successful cochlear implantation.(30)(31) **(EG 2)** A prospective cohort study comparing outcomes in 136 adolescents with severe to profound hearing loss treated with cochlear implants with outcomes in 340 adolescents with severe to profound hearing loss without cochlear implants found that children with cochlear implants had better academic performance (as measured with the Woodcock-Johnson III or IV assessment) and improved quality of life (as measured with the Pediatric Quality of Life Inventory) compared with children without implants; the largest benefits in academic performance and quality of life were seen in the subgroup of patients who received a cochlear implant before 18 months of age.(48) **(EG 2)**

For replacement of existing cochlear implant or component needed, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** Health technology assessments note that with long-term use, cochlear implants may require replacement for upgraded technology or device malfunction. Additionally, individual components (eg, headset/headpiece, microphone, transmitting coil or cable, external speech processor or controller, battery) require episodic maintenance, repair, or replacement.(2)(10) **(EG 2)**

Rationale

Use of this MCG care guideline helps the clinician determine if a particular treatment, medication, or service might be appropriate for a specific patient, taking into account their unique health complexities.

Use of these evidence-based clinical criteria to support decision making benefits the patient by identifying patient-specific complex clinical factors and conditions, promoting personalized treatment. Utilizing evidence-based clinical criteria promotes patient safety by helping ensure that potential patient benefits outweigh the risks. In addition, the use of evidence-based guidelines can increase consistency in treatment thresholds, leading to less variation in care and promoting equitable treatment among patients.

Related CMS Coverage Guidance

This guideline supplements but does not replace, modify, or supersede existing Medicare regulations or applicable National Coverage Determinations (NCDs) or Local Coverage Determinations (LCDs).

Code of Federal Regulations (CFR): 42 CFR 419.22(65); 42 CFR 422.101(66)

Internet-Only Manual (IOM) Citations: CMS IOM Publication 100-02, Medicare Benefit Policy Manual, Chapter 14 - Medical Devices(67); CMS IOM Publication 100-02, Medicare Benefit Policy Manual, Chapter 15 - Covered Medical and Other Health Services(68); CMS IOM Publication 100-02, Medicare Benefit Policy Manual, Chapter 16 - General Exclusions from Coverage(69)

Medicare Coverage Determinations: Medicare Coverage Database(70)

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Footnotes

[A] Available evidence supports the use of unilateral, sequential bilateral, or simultaneous bilateral cochlear implants in adults and children.(2)(3)(10)(11)(12)(13) [A in Context Link 1, 2, 3, 4]

[B] Pure-tone air-conduction testing assesses hearing function; the threshold is determined by presenting pure tones to a patient and determining the lowest level at which the tone can be heard. A pure-tone average threshold of 25 dB or less is considered normal hearing. Mild hearing loss is characterized by pure-tone average thresholds of 26 dB to 39 dB, while moderate hearing loss falls within the range of 40 dB to 69 dB. Severe hearing loss is characterized by pure-tone average thresholds of 70 dB to 89 dB; pure-tone average thresholds of 90 dB or greater are consistent with profound hearing loss.(36) [B in Context Link 1, 2]

[C] Implantation in infants younger than 9 months with significant hearing loss may be considered when a delay in implantation could worsen outcomes, particularly in cases of hearing loss after bacterial meningitis with evidence of cochlear ossification and fibrosis on imaging.(42)(47)(60)(61) [C in Context Link 1]

[D] Auditory function can be measured by speech, listening, or language skills utilizing instruments such as Early Speech Perception Test (ESP), Meaningful Auditory Integration Scale (MAIS), or Lexical Neighborhood/Multisyllabic Lexical Neighborhood Tests (LNT-MLNT).(3)(64) [D in Context Link 1]

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